Identification of liquid organic compounds

organic compounds
(contain carbon and hydrogen)

- open chain aliphatic
  - Straighchain
    $CH_3-CH_2-CH_2-CH_3$ chain
    Butane
  - Branched
    $CH_3-CH(CH_3)-CH_3$
    2-methyl Propane
- cyclic compounds
  - Homocyclic compounds
    - aliphatic compound
    - aromatic compound
  - Heterocyclic compounds:

* classification of organic compounds according to functional groups :-

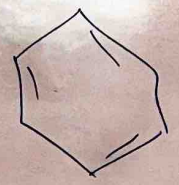
1) Planar
2) conjugation
3) Huck Rule

$4n+2 = \pi e$

$4n+2 = 6$

$4n = 4$

$n = 1$



Keto

aliph

acet

CH_3-

5) co

aliph

CH

ac

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ydrogen)

ic compounds

Heterocyclic compounds :-

c)

yclic compound

aromatic compound

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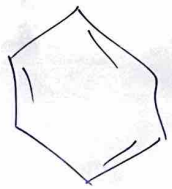
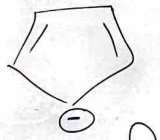
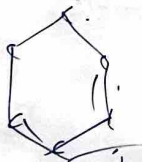






cyclohexane

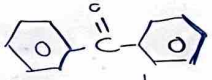
4) Keto group $R-\overset{O}{\overset{\|}{C}}-R'$ $R' \neq H$

aliphatic

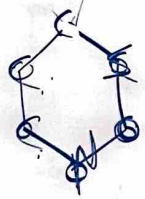
acetone

$CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

aromatic



Benzophenone :-



5) carboxylic group [R-COOH]

aliphatic

CH_3COOH

acetic acid

aromatic

COOH

Benzoic acid

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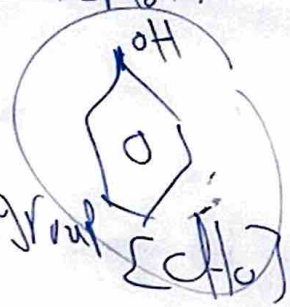
1] Hydroxyl group [OH]

aliphatic alcohols
Ethanol
CH_3-CH_2-OH

aromatic Phenols



2] aldehyde group [CHO] $R-\overset{O}{\overset{\|}{C}}-H$

aliphatic
CH_3CHO
acetaldehyde

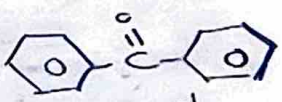
aromatic

Benzaldehyde

Keto group $R-\overset{O}{\overset{\|}{C}}-R'$ $R' \neq H$

aliphatic
acetone
$CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

aromatic



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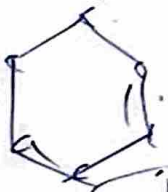
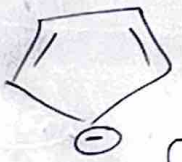


cyclohexane



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4) Salts of carboxylic acids

Sodium salts

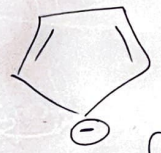
$R\text{-}COOH + NaOH \rightarrow R\text{-}COONa + H_2O$

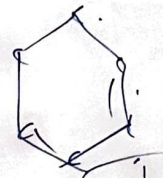
ammonium salts

$R\text{-}COOH + NH_4OH \rightarrow R\text{-}COONH_4 + H_2O$

3) amines ⇒ [$R\text{-}NH_2$]

3) carbohydrates: $C_nH_{2n}O_n$





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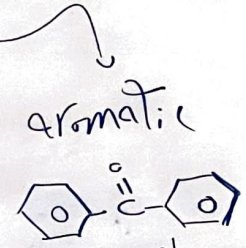

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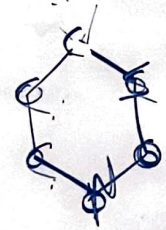

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